Measures of Central Tendency

Measure	Individual Series	Discrete Series	Continuous Series
Mean	Direct: $\bar{X} = \Sigma X / N$ Short-cut: $\bar{X} = A + \Sigma d/N$ A= Assumed mean N=Number of observations $d = (X - A)$	Direct: $\bar{X} = \Sigma fX / N$ Short-cut: $\bar{X} = A + \Sigma fd/N$ A= Assumed mean N=Σf = Total frequency $d = (X - A)$	Direct: $\bar{X} = \Sigma fm / N$ Short-cut: $\bar{X} = A + \Sigma fd / N$ Step-deviation: $\bar{X} = A + (\Sigma fd'/N) \times i$ A= Assumed mean N=Σf = Total frequency *i = Class interval* *m = mid-point of class (lower limit + higher limit /2)* $d = (m - A)$ $d' = (m-A)/i$
Combined Mean	$\bar{X}_{12} = (N_1\bar{X}_1 + N_2\bar{X}_2) / (N_1+N_2)$		
Weighted Mean	$\bar{X}w = \Sigma WX / \Sigma W$		
Median	$(N+1)/2^{th}$ item N=Number of observations Arrange data in ascending order Odd N — median is an actual item Even N — average the two middle items.	$(N+1)/2^{th}$ item Arrange the data in ascending order of magnitude N=Σf = Total frequency	i. Median class= $N/2^{th}$ item ii. Median value= $L + [(N/2 - c.f.)/f] \times i$ *Arrange in ascending order* N=Σf = Total frequency L= *lower limit of* Median class *c.f. = Cumulative freq. before median class* *f = Actual frequency of* Median class *i = Class interval*
Mode	Value occurring most frequently	Value with highest frequency	Mode = $L + [\Delta_1 / (\Delta_1+\Delta_2)] \times i$ $\Delta_1 = \lvert f_1 - f_0 \rvert \quad \Delta_2 = \lvert f_1 - f_2 \rvert$ f_1 = Class with maximum frequency L= *lower limit of* f_1 class *i = class interval* f_0, f_2 *freq. of preceding, and succeeding class of* f_1

Quartile	$Q_1 = (N+1)/4$ $Q_3 = 3(N+1)/4$ N=Number of observations	$Q_1 = (N+1)/4$ $Q_3 = 3(N+1)/4$ N=Σf = Total frequency	Qk = L + (kN/4 − cf)/f *i N=Σf = Total frequency *k=Quartile Number* Q_k Class= kN/4 L=Lower boundary of Q_k class cf = Cumulative frequency of previous class of Q_k Class f = Normal frequency of Q_k class i = Class interval
Decile	$D_k = k(N+1)/10$ N=Number of observations	$D_k = k(N+1)/10$ N=Σf = Total frequency	D_k=L + (kN/10 − cf)/f *i N=Σf = Total frequency *k=Decile Number* D_k Class= kN/10 L: Lower boundary of D_k class cf = cumulative frequency of previous class of D_k Class f = Normal frequency of D_k class i = Class interval
Percentile	$P_k = k(N+1)/100$ N=Number of observations	$P_k = k(N+1)/100$ N=Σf = Total frequency	P_k =L + (kN/100 − cf)/f *i N=Σf = Total frequency *k=Percentile Number* P_k Class= kN/100 L: Lower boundary of P_k class cf = Cumulative frequency of previous class of P_k Class f = Normal frequency of P_k class i = Class interval

Measures of Spread or Dispersion

Measure	Individual Series	Discrete Series	Continuous Series
Range	$L - S$ L = Value of the largest item S = Value of the smallest item	$L - S$ L = Value of the largest item S = Value of the smallest item	$L - S$ L = Upper limit of highest class S = Lower limit of lowest class
Coefficient of Range	$(L - S) / (L + S)$ L = Value of the largest item S = Value of the smallest item	$(L - S) / (L + S)$ L = Value of the largest item S = Value of the smallest item	$(L - S) / (L + S)$ L = Upper limit of highest class S = Lower limit of lowest class
Interquartile Range	$Q_3 - Q_1$ Use individual series formula calculate Q_3 & Q_1	$Q_3 - Q_1$ Use Discrete series formula to calculate Q_3 & Q_1	$Q_3 - Q_1$ Use Continuous series formula to calculate Q_3 & Q_1
Quartile Deviation	$(Q_3 - Q_1)/2$ Use individual series formula calculate Q_3 & Q_1	$(Q_3 - Q_1)/2$ Use Discrete series formula to calculate Q_3 & Q_1	$(Q_3 - Q_1)/2$ Use Continuous series formula to calculate Q_3 & Q_1
Coefficient of Q.D.	$(Q_3 - Q_1)/ (Q_3 + Q_1)$ Use individual series formula calculate Q_3 & Q_1	$(Q_3 - Q_1)/ (Q_3 + Q_1)$ Use Discrete series formula to calculate Q_3 & Q_1	$(Q_3 - Q_1)/ (Q_3 + Q_1)$ Use Continuous series formula to calculate Q_3 & Q_1
Standard Deviation(σ)	Actual Mean Method $\sigma = \text{sqrt}\ (\Sigma x^2 / N)$ *where* $x = (X - \bar{X})$ N=Number of observations	Actual Mean Method $\sigma = \text{sqrt}\ (\Sigma fx^2 / N)$, $x = (X - \bar{X})$ $N=\Sigma f$ = Total frequency	$\sigma = \text{sqrt}\ (\Sigma fd^2/N - (\Sigma fd/N)^2) \times i$ m = mid-point i = class interval A = assumed mean $d = (m - A) / i$ $N=\Sigma f$ = Total frequency
	Assumed Mean Method $\sigma = \text{sqrt}\ [\Sigma d^2/N - (\Sigma d/N)^2]$ *where* $d = (X - A)$ *A: Assumed Mean* *N: No. of observations*	Assumed Mean Method $\sigma = \text{sqrt}\ (\Sigma fd^2/N - (\Sigma fd/N)^2)$, $d = (X - A)$ $N=\Sigma f$ = Total frequency	
		Step Deviation Method $\sigma = \sqrt{(\ \Sigma fd^2/N - (\Sigma fd/N)^2\ )} \times i$, $d = (X-A)/i$ i = class interval $N=\Sigma f$ = Total frequency	

Measures of Distribution

<table>
<tr><th>Measure</th><th>Individual Series</th><th>Discrete Series</th><th>Continuous Series</th></tr>
<tr><td>Karl Pearson's Coeff.</td><td>Sk= [$\bar{X}$ − Mode] / σ
Sk= 3($\bar{X}$ − Med.) / σ (if series is bimodal)

Use formulas of Individual series to calculate

1. $\bar{X}$ = Mean
2. Mode = Value occurring most frequently
3. σ = Standard Deviation</td><td>Sk= [$\bar{X}$ − Mode] / σ
Sk= 3($\bar{X}$ − Med.) / σ (if series is bimodal)

Use formulas of Discrete series to calculate

1. $\bar{X}$ = Mean
2. Mode = Value with highest frequency
3. σ = Standard Deviation</td><td>Sk= [$\bar{X}$ − Mode] / σ
Sk= 3($\bar{X}$ − Med.) / σ (if series is bimodal)

Use formulas of Continuous series to calculate

1. $\bar{X}$ = Mean
2. Mode = Value with highest frequency
3. σ = Standard Deviation</td></tr>
<tr><td></td><td colspan="3"><table><tr><th>Distribution</th><th>Order of Mean, Median, Mode</th><th>Relative Value</th></tr><tr><td>Positively Skewed</td><td>Mode < Median < Mean</td><td>Mean is highest, Mode is lowest</td></tr><tr><td>Negatively Skewed</td><td>Mean < Median < Mode</td><td>Mode is highest, Mean is lowest</td></tr></table></td></tr>
<tr><td>Bowley's Coefficient</td><td>= (Q_3 + Q_1 − 2 × Median) ÷ (Q_3 − Q_1)
Use individual series formula calculate Q_3 & Q_1</td><td>= (Q_3 + Q_1 − 2 × Median) ÷ (Q_3 − Q_1)
Use Discrete series formula to calculate Q_3 & Q_1</td><td>= (Q_3 + Q_1 − 2 × Median) ÷ (Q_3 − Q_1)
Use Continuous series formula to calculate Q_3 & Q_1</td></tr>
<tr><td>Kurtosis — β2</td><td>β2 = μ4 / μ2²

$\bar{x}$= Σx / n
μ_2 = Σ(x− $\bar{x}$)² / n
μ_4 = Σ(x− $\bar{x}$)⁴ / n</td><td>β2 = μ4 / μ2²

$\bar{x}$ = Σfx / N
μ_2 (σ^2) = Σf(x−$\bar{x}$)² / N
μ_4 = Σf(x−$\bar{x}$)⁴ / N</td><td>β2 = μ4 / μ2²

$\bar{x}$ = Σfx / N
μ_2 = Σf(x−$\bar{x}$)² / N
μ_4 = Σf(x−$\bar{x}$)⁴ / N</td></tr>
<tr><td></td><td colspan="3"><table><tr><th>Value of β2</th><th>Interpretation</th><th>Shape</th><th>Meaning</th></tr><tr><td>β2 = 3</td><td>Mesokurtic</td><td>Narrow, tall peak</td><td>More peaked than normal</td></tr><tr><td>β2 > 3</td><td>Leptokurtic</td><td>Normal curve</td><td>Same peakedness as normal</td></tr><tr><td>β2 < 3</td><td>Platykurtic</td><td>Broad, flat top</td><td>Less peaked than normal</td></tr></table></td></tr>
</table>

Multivariate Exploration

Pearson's Coefficient of Correlation

METHOD 1: Deviations Taken from Actual Mean

$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)}$ OR $r = \Sigma xy / (N \sigma x \sigma y)$

METHOD 2: Direct Method — Without Taking Deviations

$r = [N \Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[N \Sigma X^2 - (\Sigma X)^2] \times [N \Sigma Y^2 - (\Sigma Y)^2]}$

METHOD 2: Deviations Taken from an Assumed Mean

$r = [N \Sigma dx\, dy - (\Sigma dx)(\Sigma dy)] / \sqrt{[N \Sigma dx^2 - (\Sigma dx)^2] \times [N \Sigma dy^2 - (\Sigma dy)^2]}$

Perfect Positive	r = +1
Perfect Negative	r = −1
High Degree Positive	0.75 to +0.99
High Degree Negative	−0.75 to −0.99
Low Degree Positive	0 to +0.5
Low Degree Negative	0 to −0.5
No Correlation	r ≈ 0

Univariate Visualization

Box Plot	• Min / Max • Q1 (Lower Quartile) • Q2 (Median) • Q3 (Upper Quartile) Where Q1, Q2 and Q3 are calculated for individual series • IQR = Q3 − Q1 • Lower Limit = Q1 − 1.5×IQR • Upper Limit = Q3 + 1.5×IQR • Outliers: Data < Lower Limit & data>Upper Limit	• Min / Max • Q1 (Lower Quartile) • Q2 (Median) • Q3 (Upper Quartile) Where Q1, Q2 and Q3 are calculated for Discrete series • IQR = Q3 − Q1 • Lower Limit = Q1 − 1.5×IQR • Upper Limit = Q3 + 1.5×IQR • Outliers: Data < Lower Limit & data>Upper Limit	• Min / Max • Q1 (Lower Quartile) • Q2 (Median) • Q3 (Upper Quartile) Where Q1, Q2 and Q3 are calculated for Continuous series • IQR = Q3 − Q1 • Lower Limit = Q1 − 1.5×IQR • Upper Limit = Q3 + 1.5×IQR • Outliers: Data < Lower Limit & data>Upper Limit

Distribution Chart	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$ [For individual Series calculate μ (Mean) and σ] • x = value of the variable • μ = mean (central value) • σ = standard deviation • σ² = variance • e = Euler's constant (≈ 2.718) • π = 3.1416 **68.2% of data within ±1σ** **95.4% of data within ±2σ** **99.7% of data within ±3σ**	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$ [For Discrete Series calculate μ (Mean) and σ] • x = value of the variable • μ = mean (central value) • σ = standard deviation • σ² = variance • e = Euler's constant (≈ 2.718) • π = 3.1416 **68.2% of data within ±1σ** **95.4% of data within ±2σ** **99.7% of data within ±3σ**	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$ [For Continuous Series calculate μ (Mean) and σ] • x = value of the variable • μ = mean (central value) • σ = standard deviation • σ² = variance • e = Euler's constant (≈ 2.718) • π = 3.1416 **68.2% of data within ±1σ** **95.4% of data within ±2σ** **99.7% of data within ±3σ**